

Version-Aware Normalization of Findings Against MITRE ATT&CK v19

Typed Data Models and Revocation-Resolving Technique Lookup for Security Tooling

github.com/poojakira/attack-v19-core - public, default branch main.

EVIDENCE POSTURE

Verified, scoped, reproducible

Claims are constrained to the repository evidence snapshot and stated limitations.

165

Tests passed

63.83%

Statement coverage

113

Test functions

11

Test files

SYSTEM VIEW

Architecture and Threat-Control Flow

Trust Boundary

Input surface

Data, models, requests, policies, or artifacts entering the tested path.

Control surface

Repository-specific validation, analysis, and fail-closed checks.

Evidence surface

Findings, metrics, tests, logs, or reproducible artifacts.

This diagram describes the tested repository path, not a universal deployment guarantee.

01

Download/validate ATT&CK STIX data with SHA-256 verification.

02

Parse techniques/tactics/platform metadata into typed structures.

03

Resolve deprecated/revoked IDs through the v19 remap table.

04

Build in-memory indexes for ID, tactic, platform, and keyword lookup.

05

Regression-test migration and data-loader behavior across supported Python versions.

Research Contribution

A typed ATT&CK v19 data layer that normalizes findings across version churn by resolving revoked/renamed technique IDs through a maintained remap table over SHA-256-verified STIX data, with indexed lookup for downstream security tooling.

Verified Evidence

EV-01	165 tests passed.	EV-02	63.83% statement coverage; CI gate is 60%.
EV-03	113 test functions across 11 test files; parametrization expands these to 165 executed tests.	EV-04	Current repository mapping evidence retains:
EV-05	22 revoked technique IDs	EV-06	Revoked technique IDs: 22

Method and Control Path

01	Download/validate ATT&CK STIX data with SHA-256 verification.
02	Parse techniques/tactics/platform metadata into typed structures.
03	Resolve deprecated/revoked IDs through the v19 remap table.
04	Build in-memory indexes for ID, tactic, platform, and keyword lookup.
05	Regression-test migration and data-loader behavior across supported Python versions.

Limitations and Residual Risk

- No additional limitation text was found in the research brief. Consult the README before generalizing results.

Reproducibility

Expected current CI evidence: 165 passed, 63.83% coverage.

Evidence Provenance

Claim discipline
Metrics come from the repository's verified metrics and research brief.
Scope discipline
Benchmarks are reported only at their documented fixture, dataset, or test scope.
Reproduction
PDF and PNG are generated locally from committed evidence without paid services.